

- 3** The topshell is a marine mollusc widely found on European coasts.
It lives on rocky shores and is found in rock pools, on bare rocks and amongst seaweed.

The photograph shows a group of topshells.



Scale 
1 cm

(Source: © Nature Picture Library / Alamy Stock Photo)

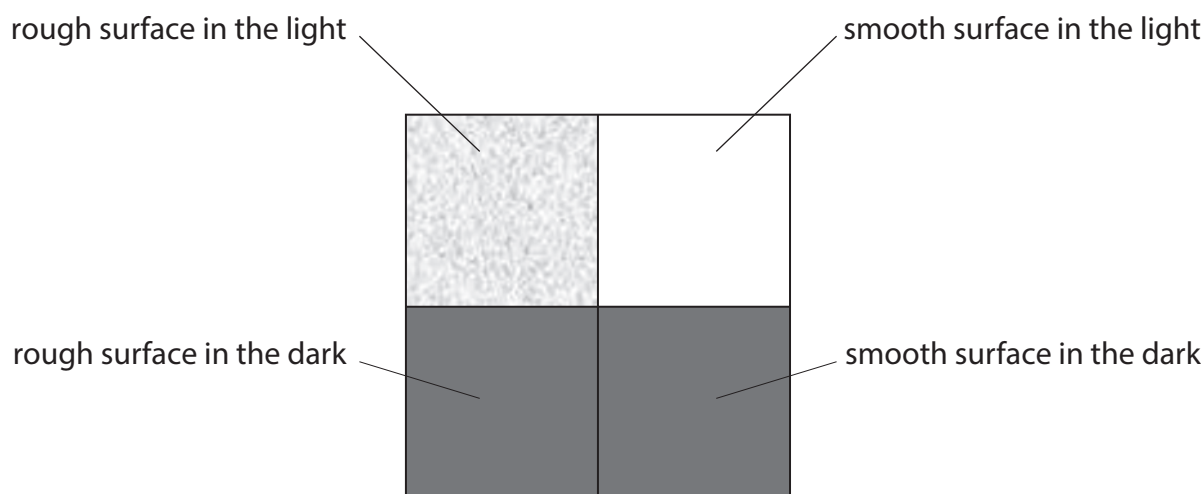
- (a) Calculate the magnification of this photograph.

(1)

Answer

(b) A student investigated the types of habitat preferred by these topshells.

The diagram shows a tray with four different environmental conditions used by this student.



The student suggested the null hypothesis that topshells show no significant habitat preference.

Five topshells were placed in each of the four areas and the tray was left for five minutes for the animals to move around.

The number of topshells in each area was then recorded.

This was repeated three more times and the results totalled.

The table shows the results of this investigation.

Area	Number of topshells in each area of the tray after 5 minutes				Total number observed in each area of the tray O	Total number expected in each area of the tray E
	Trial 1	Trial 2	Trial 3	Trial 4		
Smooth surface in the light	3	2	4	3		20
Rough surface in the light	4	4	4	3		20
Smooth surface in the dark	6	7	6	5		20
Rough surface in the dark	7	7	6	9		20

(i) Calculate the Chi-squared value using the formula:

(3)

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Answer

(ii) The table shows some critical values for Chi-squared at different degrees of freedom.

Degrees of freedom	<i>p</i> value = 0.050
1	3.841
2	5.991
3	7.815
4	9.488

Comment on the results of this investigation.

Use the null hypothesis, your calculated Chi-squared value and the table of critical values to support your answer.

(3)

[illegible]

(iii) Justify **three** improvements that could be made to this investigation.

(3)

[illegible]

(Total for Question 3 = 10 marks)